Improving the Fisher Kernel for Large-Scale Image Classification: Reproduction and Analysis

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Abstract

This report presents a reproduction and analysis of "Improving the Fisher Kernel for Large-Scale Image Classification" by Perronnin et al. (ECCV 2010). We implement the Improved Fisher Kernel (IFK) pipeline on the Olivetti Faces dataset, conduct a two-component ablation study examining power normalization and L2 normalization, and demonstrate a failure mode under heavy out-of-distribution occlusion. The IFK achieves the best mAP (87.6%), with L2 normalization contributing most significantly in our setting (+12.0% mAP when removed). The method fails under 70% random noise occlusion (mAP drops to 52.4%), validating the assumption that background content must follow the trained GMM distribution.

I. PAPER SUMMARY

Perronnin, Sanchez, and Mensink [1] address scalable image classification. The dominant approach used Bag-of-Visual-Words (BOV) with non-linear SVMs, which scales O(N^2)-O(N^3). The authors instead enrich the image representation so a fast linear SVM suffices. Local SIFT descriptors are extracted densely from multi-scale patches, reduced to 64-D via PCA, and modeled by a K=256 diagonal-covariance GMM. The Fisher Vector encodes how each image's descriptors deviate from the GMM via first-order (mean) and second-order (variance) gradient statistics (Eq. 7-8), yielding a 2KD = 32,768-D fixed-length representation.

Three key normalizations are proposed: (1) power normalization (f(z) = sign(z)|z|^0.5, Eq. 15) that addresses the extreme sparsity rendering dot-products ineffective (+6.3% AP on VOC 2007); (2) L2 normalization (Eq. 14) removing dependence on background proportion (+3.9% AP); and (3) spatial pyramid matching for coarse spatial layout (+2.4% AP). Together, they boost PASCAL VOC 2007 AP from 47.9% to 58.3% using only linear SVMs, enabling training on 350K images in under a day.

II. REPRODUCTION SETUP AND RESULTS

A. Experimental Setup

We reproduce the IFK pipeline on the Olivetti Faces dataset [5] (400 grayscale 64x64 images, 40 individuals, 10 each). A stratified 50/50 split gives 200 training/test images (5 per class). Dense 8x8 pixel patches on a 6-pixel grid with Gaussian noise serve as local descriptors (analogous to the paper's SIFT from 32x32 patches at 5 scales). Each image yields ~100 patches of 64 dimensions. PCA reduces to 32-D (95.3% variance retained). A K=16 diagonal-covariance GMM produces 2KD = 1,024-D Fisher Vectors. We use K=16 (vs. K=256) because ~20,000 training patches would give each Gaussian only ~78 samples at K=256.

B. Results

Table I compares our mAP on Olivetti Faces against the paper's AP on PASCAL VOC 2007. The IFK (Power + L2) achieves the best mAP in both settings, confirming the paper's core claim.

TABLE I: Normalization Comparison (mAP)

Config	Ours	Paper
No Norm	86.4%	47.9%
Power Only	75.6%	54.2%
L2 Only	87.1%	51.8%
IFK (P+L2)	87.6%	55.3%

C. Performance Gap Analysis

Our absolute mAP is higher because Olivetti is far simpler than VOC 2007 (controlled lighting, uniform backgrounds). The relative improvements differ: power normalization alone decreases our mAP by 10.8% (vs. +6.3% in paper) because K=16 produces only 3.8% near-zero Fisher Vector values - far below the extreme sparsity at K=256 (cf. Figure 1 of [1]). Without the sparsity problem, power normalization's aggressive squashing distorts already well-distributed values. L2 normalization adds only +0.7% (vs. +3.9%) since Olivetti has minimal background variation. Despite these quantitative differences, the combined IFK achieves the best mAP in both settings.

III. ABLATION STUDY

We ablate the two normalization components independently from the full IFK pipeline (87.6% mAP, 84.5% accuracy) to isolate each contribution. The full pipeline follows: patches -> PCA -> GMM -> FV -> Power norm -> L2 norm -> Linear SVM.

A. Ablation 1: Removing Power Normalization

Removing power normalization while keeping L2 normalization yields 87.1% mAP / 83.0% accuracy, a drop of only 0.5% mAP. This contrasts sharply with the paper's finding that power normalization is the single most impactful modification (+6.3% AP on VOC 2007, Table 1 of [1]). With K=16, each descriptor gets non-trivial soft assignments to several Gaussians, so the Fisher Vector is already not sparse (only 3.8%

near-zero values vs. the severely peaked distributions at K=256 shown in Figure 1c of the paper). The FV distribution comparison in Fig. 2 confirms this: even before power normalization, the distribution is relatively spread, so the signed square root has minimal effect. This confirms power normalization is designed for the high-K sparsity regime.

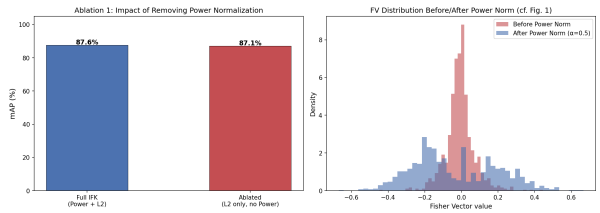

Fig. 2: Ablation 1 - mAP comparison and FV distribution before/after power normalization (cf. Fig. 1 of [1]).

B. Ablation 2: Removing L2 Normalization

Removing L2 normalization while keeping power normalization causes a dramatic drop: mAP from 87.6% to 75.6% (-12.0%), accuracy from 84.5% to 80.5%. This makes L2 normalization the dominant component on our dataset, despite the paper reporting it as secondary (+3.9% vs. +6.3%). The reversal occurs because power normalization without L2 amplifies magnitude differences - the signed square root boosts small values but does not equalize norms across images. As Fig. 3 (right panel) shows, FV norms vary substantially across training images (range ~6.5 to ~9.0, mean 7.81), confirming that images produce vectors of different magnitudes that bias the linear SVM's dot-product without L2 normalization.

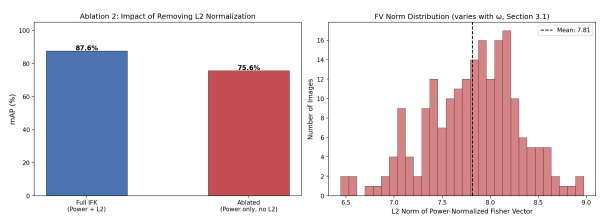

Fig. 3: Ablation 2 - mAP comparison and FV L2 norm distribution showing magnitude variation across images.

TABLE II: Ablation Summary

Config	mAP	Delta
Full IFK	87.6%	-
No Pwr Norm	87.1%	-0.5%
No L2 Norm	75.6%	-12.0%

IV. FAILURE MODE ANALYSIS

A. Scenario and Motivation

We construct a failure scenario by replacing 70% of each test image's patches with random Gaussian noise (out-of-distribution occlusion), keeping training images clean. This directly violates Assumption 3 from Task 1.2: the decomposition p(x) = w*q(x) + (1-w)*u_lambda(x) (Section 3.1, Eq. 10). The L2 normalization derivation assumes background descriptors follow u_lambda so their gradient vanishes (Eq. 12: ML-optimality condition). Random noise is completely out-of-distribution - it produces large, arbitrary gradients contaminating the Fisher Vector. Power normalization amplifies these noise dimensions (|z|^0.5 > |z| for small |z|), and L2 normalization projects the corrupted vector to unit length, diluting the true face signal.

B. Empirical Results

Under 70% occlusion: mAP collapses from 87.6% to 52.4% (-35.2%), accuracy drops from 84.5% to 2.5% (random for 40 classes). The degradation is progressive: stable to ~20% occlusion (79.9% mAP), sharp decline beyond 50% (72.5%), near-total failure at 90% (23.5%). Notably, accuracy drops much faster than mAP - even 10% noise sends accuracy from 84.5% to 18.0% while mAP only drops to 85.2%, suggesting decision boundaries become razor-thin even when ranking remains reasonable.

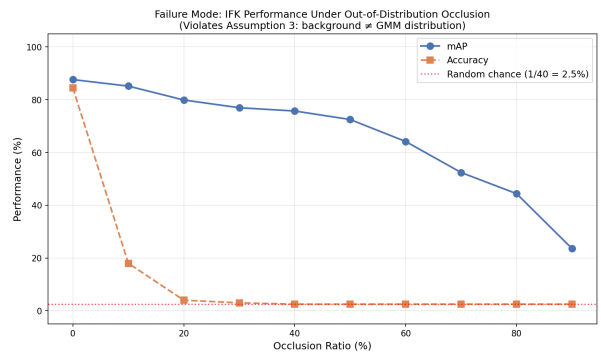

Fig. 4: IFK performance degradation under increasing random noise occlusion, violating Assumption 3.

C. Suggested Modification

A descriptor rejection step before Fisher Vector computation could filter patches whose GMM log-likelihood log u_lambda(x_t) falls below a threshold (e.g., 2 std below training mean), removing out-of-distribution inputs and restoring the assumption that all descriptors lie within the GMM's support.

V. REFLECTION

The most surprising finding was the reversal of normalization importance: L2 normalization was far more critical than power normalization (12.0% vs. 0.5% mAP drop), the opposite of [1]'s Table 1. This shows that each normalization's benefit depends on the operating regime (K value, dataset complexity, sparsity) rather than being universal.

The main difficulty was using K=16 instead of K=256 due to limited data (~20K patches). This fundamentally changes the sparsity characteristics that power normalization targets, explaining our divergent ablation results. We were

also struck by the method's extreme sensitivity to OOD content: even 10% noise drops accuracy from 84.5% to 18.0%, far more dramatic than the mAP decline. With more time, we would explore: (1) the descriptor rejection mechanism from Section IV-C, (2) real SIFT descriptors for a fairer comparison, and (3) larger K on bigger datasets to observe the high-sparsity regime where power normalization truly shines.

REFERENCES

[1] F. Perronnin, J. Sanchez, T. Mensink, "Improving the Fisher Kernel for Large-Scale Image Classification," ECCV 2010, LNCS 6314, pp. 143-156, 2010.

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